Lesson 4-3 · Period 1

Rational Expressions — Equivalent & Simplify

Klimsara-adapted · Student-centered · No Blooket

DOK 1 5 min *notice & wonder*

Graph of $y = x + 2$Consider the graph of the function $y = x + 2$.

1. What is the domain of this function?
2. Sketch a version that **excludes** $x = 2$ from the domain.
3. What kind of function might have a graph like that? Explain.

Write 1 **notice** + 1 **wonder** on your packet. One sentence: what is the equation doing at that skipped x-value?

DOK 1–2 | 12 min | *teacher models*

Equivalent + Simplify Rules (keep on your page)

1. **Factor** the numerator AND denominator completely.
2. Cancel only common **FACTORS** — never terms across $+$ or $-$.
3. State domain restrictions: every zero of the **ORIGINAL** denominator is excluded — including factors you canceled.

Common Error: Do NOT cancel terms that are added or subtracted across a fraction bar. Factor completely FIRST, then cancel only identical factor

DOK 1–2 | 33 min | *student work*

7 items in order. Plan ∼4–5 min per item. Cut #19 if pace slows.

Try It 1 · Equivalent expression Write an expression equivalent to $(x-4)/x$ over domain $\{x \mid x \neq 0 \text{ or } -2\}$.

Practice #20 · Equivalent over restricted domain Write an equivalent expression: $x(x-2)/(x-5)$ for all x except 5 and -6.

Practice #22 · Simplify to domain Simplify $(x^2$

Explore (continued)

DOK 1–2 | 33 min | *student work*

Try It 2 · Simplify by factoring and canceling Simplify each expression and state the domain.

a. $(x^2 + 2x + 1)/(x^3 - 2x^2 - 3x)$ **b.** $(x^3 + 4x^2 - x - 4)/(x^2 + 3x - 4)$

Practice #17 · Construct Arguments Explain how you can tell whether a rational expression is in simplest form.

Share / Summary

DOK 2 | 5 min | *academic conversation*

Sentence frame · Domain & Simplify "The factored form is ________, so the domain excludes $x =$ _____. After canceling the factor _______, the simplified expression is _______ with the **same** domain restriction."

Bridge to Period 2 Next class: multiply and divide rational expressions.

The same factor-and-cancel logic applies — plus you'll work

Exit Ticket — Summary Recap

Algebra 2 · Lesson 4-3 · Period 1

DOK 1–2 | 2 min | *summary recap — not graded*

Complete on your exit ticket To simplify $(x^2 + 3x - 10)/(x^2 - 4)$ I first factor to ________________.

The restricted domain values are $x \neq$ __________.

After canceling the factor __________, my simplified expression is __________.

One biggest thing I learned today: ________________________.